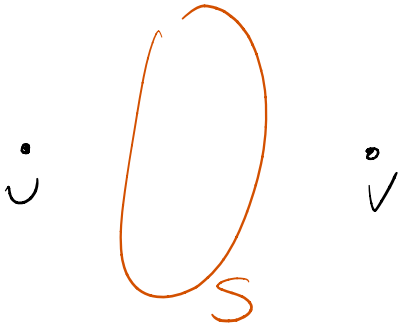
Recall: Let G be a graph



- A **vertex-cut** is a subset $S \subset V(G)$ such that $G - S$ is disconnected.
- The **vertex-connectivity** $\kappa(G)$ is the minimum cardinality of a vertex-cut.
- For $G = K_n$, we set $\kappa(G) = n - 1$ (because it contains no vertex-cuts).
- If k is a non-negative integer, we say that G is k-connected if $\kappa(G) \geqslant k$.

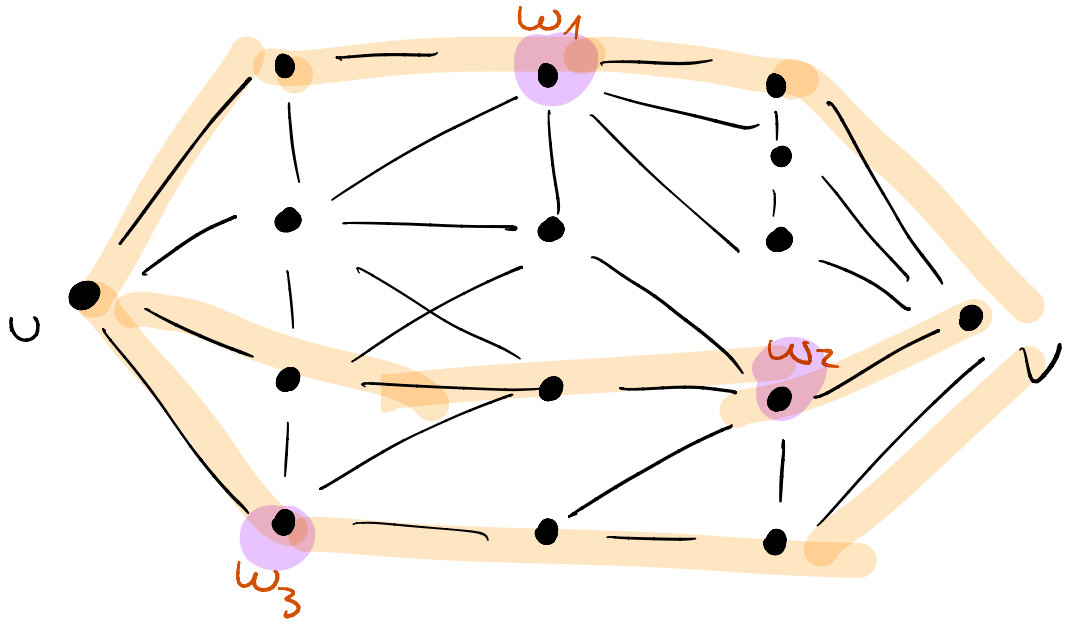
Today: computing $\kappa(G)$ using paths.

Definition. *Fix $u, v \in V(G)$.*

- *A u–v **separating set** is a vertex cut $S \subset V(G)$ such that u and v belong to different components of $G - S$.*
- *A **minimum** u–v **separating set** is a u–v separating set with minimum cardinality.*
- *A collection $\{P_1, \ldots, P_k\}$ of u–v paths is **internally disjoint** if every two of these paths have no vertices in common other than u and v.*

Note: If S is a minimum u–v separating set, there are **at most** $|S|$ internally disjoint u–v paths.
Next: actually there are exactly $|S|$.

Example:



$\{w_1, w_2, w_3\}$ is a minimum u–v separating set

Thus we have exactly 3 internally disjoint u–v paths

Theorem (Menger's theorem 5.16). *Let* u *and* v *nonadjacent vertices in a graph* G. *The minimum number of vertices in a* u–v *separating set equals the maximum number of internally disjoint* u–v *paths in* G.

Proof : By induction on $m = |E(G)|$

Trivially True for $m = 0$

Fix G with $|E(G)| = m > 0$.

Let $u, v \in V(G)$ such That $uv \notin E(G)$

Let K denote The size of a minimum u–v separating set.

Want To show G contains K internally disjoint u–v paths.

Clear if $K = 0, 1$. Assume $K \geq 2$

($K = 1$: 

)

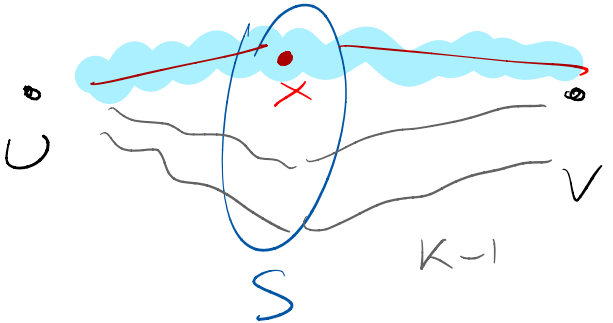
Case 1: Some minimum u–v separating set S contains a vertex x adjacent to both u and v.

The graph $G-x$ has $|E(G-x)| < m$

Also $S-\{x\}$ is a minimum u–v separating set in $G-x$

By IH, in $G-x$ we have $k-1$ $(=|S-\{x\}|$ internally disjoint u–v paths.

Add the path (u, x, v) in G to get k internally disjoint.



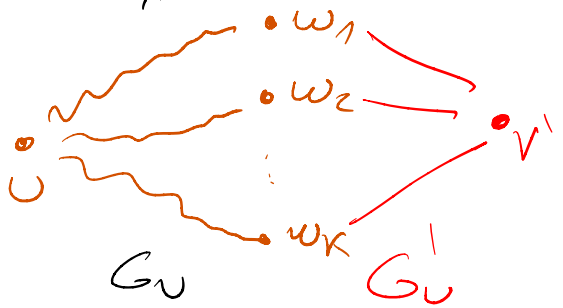
Case 2: Some minimum u–v separating set S contains a vertex non-adjacent to u and a vertex non-adjacent to v.

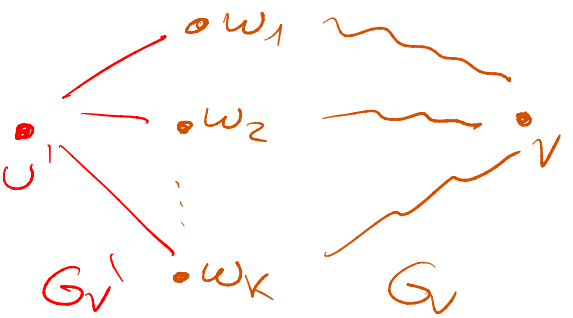
Let $S = \{w_1, \ldots, w_k\}$

Let G_u denote the subgraph of G consisting of all u–w_i paths with no internal vertices in S

Let G_u' graph obtained from G_u by adding new vertex v' and edges $v'w_i$, $1 \leq i \leq k$

Similarly, define G_v, v', G_v'





Note 1: G_v' and G_v' have $< m$ edges.

Since S is a separating set, in G_v we have at least the $k-$ edges we are adding to form G_v'

But since some w_i is not a neighbor of v, in G we have more edges

Note 2: S is minimum $u-v'$ separating set in G_v'

S is minimum $u'-v$ sep set in G_v'

By IH, G_v' contains k internally disjoint $u-v'$ paths $(u, \ldots, w_i, v')$ P_i

$u'-v$ $(u', w_{i1}, \ldots, v)$ Q_i

Thus P_i followed by Q_i, $1 \leq i \leq k$, give k internally disjoint $u-v$ paths in G.

Case 3: For every minimum u–v separating set S, either every vertex in S is adjacent to u and not v, or every vertex in S is adjacent to v and not u.

Let $P = (u, x, y, \ldots, v)$ be a u-v geodesic in G.

Let $e = xy$

Claim: In $G-e$, any minimum u-v sep. set has k elements

Such set has at least $k-1$ elements.

So if the claim is not true there is a u-v sep set in $G-e$

$Z = \{z_1, \ldots, z_{k-1}\}$

Then $Z \cup \{x\}$ is a minimum sep set in G. Thus every $z_i \in Z$ is a neighbor of u ($\to ux \in E(G)$)

But also $Z \cup \{y\}$ is a minimum sep set in $G \Rightarrow y$ is a neighbor of u.

↘ Contains u-neighbors

This contradicts $P = (u, x, y, \ldots)$ geodesic.

By IH, $G-e$ contains k internally disjoint u-v paths

Thus also G. ∎

Theorem (Withney's theorem 5.17). *Let* G *be a non-trivial graph and* $k \geqslant 2$ *an integer.*
G *is* k*-connected iff there are at least* k *internally disjoint* u–v *paths in* G *for each pair* u, v *of distinct vertices.*

Proof: For complete graphs ✓

⇒) Assume G is k-connected, let $u \neq v \in V(G)$

case 1: $uv \notin E(G)$. Let S be a minimum u–v sep. set.

Then $|S| \geqslant K(G) \geqslant k$, By Menger G contains at least

k internally disjoint u–v paths.

Case 2: $e = uv \in E(G)$. Hence G-e is k-1 connected → (HW 7.3)

Let S minimum sep set in G-e. Thus $|S| \geqslant K(G-e) \geqslant k-1$

By Menger, G-e contains k-1 internally disjoint u–v paths

So in G we have K (add $e = uv$)

($\Leftarrow$) Assume G contains K int. disjoint u-v paths.

Let U be a vertex cut, ie: $|U| = \kappa(G)$

Let x, y in different components of $G - U$

Since there are K internally disjoint x-y paths in G and U is x-y separating, Menger $\Rightarrow |U| \geq K$

Thus $\kappa(G) = |U| \geq K$ ▨

Exercise. *Verify that* $\kappa(H_{r,n}) = r$ *by showing that every two vertices* u, v *in* $H_{r,n}$ *are connected by* r *internally disjoint* u–v *paths for the following values of* r *and* n*: (a)* $r = 3$ *and* $n = 8$*. (b)* $r = 4$ *and* $n = 8$*.*

Theorem (5.18)**.** *Let* G *be a* k*-connected graph and let* R *be any set of* k *vertices. If a graph* H *is obtained from* G *by adding a new vertex* w *and joining* w *to the vertices on* R*, then* H *is also* k*-connected.*

Corollary (5.19)**.** *If* G *is a* k*-connected graph and* $u, v_1, v_2, \ldots, v_k$ *are distinct vertices of* G*, then there exist a family* $\{P_1, \ldots, P_k\}$ *where* P_i *is a* u-v_i *path and for* $i \neq j$*, the paths* P_i *and* P_j *have only* u *in common.*

Recall. Theorem 5.7: A graph is 2-connected iff every two vertices lie on a common cycle.

Theorem (5.20). *If* G *is a* k*-connected graph,* k ⩾ 2*, then every* k*-vertices of* G *lie on a common cycle of* G.

Idea: Grow a cycle: Let $S = \{v_1, \dots, v_k\}$

Since $k \geq 2$, G is 2-connected. Thus there is a cycle containing v_1, v_2.

Exercise 1: If there is a cycle C of length ℓ formed by $\{v_1, \dots, v_\ell\}$,

then there is a cycle containing $\{v_1, \dots, v_{\ell+1}\}$.

Exercise 2: If there is a cycle of length $> \ell$ containing $\{v_1, \dots, v_\ell\}$,

then there is a cycle containing $\{v_1, \dots, v_{\ell+1}\}$.

Upshot: If C is a cycle in G using the maximum number of elements in S, then $S \subseteq C$.

Edge-connectivity analogues:

Theorem (Menger's theorem for edge-connectivity 5.21)**.** *For distinct vertices* u, v *in a graph* G*, then*

minimum number of ***edges*** *that separate* u *and* v $=$ *maximum number of pairwise* ***edge****-disjoint* u–v *paths*

Theorem (Whitney's theorem for edge-connectivity 5.22)**.** *Let* G *be a a graph and* $k \geqslant 2$ *an integer.* G *is* k-***edge****-connected iff* G *contains* k *pairwise* ***edge-****disjoint* u–v *paths for any* $u \neq v \in V(G)$.